

LESSON 11.8

PUTTING IT ALL TOGETHER



LESSON INTRODUCTION

MA.7.AR.4.1 Determine whether two quantities have a proportional relationship by examining a table, graph, or written description.

MA.7.AR.4.2 Determine the constant of proportionality within a mathematical or real-world context given a table, graph, or written description of a proportional relationship.

MA.7.AR.4.3 Given a mathematical or real-world context, graph proportional relationships from a table, equation, or a written description.

MA.7.AR.4.4 Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

LEARNING TARGETS

- I can determine if two quantities have a proportional relationship given a real-world context.
- I can graph a proportional relationship from a table, an equation, or a written description.
- I can determine the constant proportionality of a proportional relationship.
- I can translate a given representation of a proportional relationship to a written description, table, or equation.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- How can you translate between the different representations of proportional relationships?
- How do you interpret the constant of proportionality in context?

LESSON NARRATIVE

In this lesson, students apply what they have learned about proportional relationships to real-world contexts by completing three different stations. For Station 1, students complete two different tables of values and determine if either represents a proportional relationship. Station 2 requires students to translate relationships among multiple different representations. For Station 3, students compare three different proportional relationships represented in graphs and translate them across multiple representations.

COMPONENT SUMMARY

11.8.1 WARM-UP (~5 MIN)

Connect proportional relationships to real-world examples

11.8.3 ACTIVITY (10-15 MIN)

Station 2: Compare proportional relationships from different representations

11.8.5 WRAP-UP (~5 MIN)

Reflect on learning throughout the unit thus far

11.8.2 ACTIVITY (10-15 MIN)

Station 1: Compare relationships and determine which is proportional

11.8.4 ACTIVITY (10-15 MIN)

Station 3: Compare graphs of proportional relationships in context

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Equation
- Constant of proportionality
- Unit rate
- Proportional

MATERIALS

- (Optional) Signs to label stations

ADDITIONAL PREPARATION

- Consider organizing the classroom into six stations to set up two different rotations for each of the three stations.
- Consider which groups or stations may need additional support from the teacher and keep this in mind when setting up the flow of each rotation.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse which order to divide when finding the constant of proportionality.

THINGS TO CONSIDER

- Consider how to group students for stations based on previous formative assessments.

LITERACY AND LANGUAGE ROUTINES

Take Turns

In 11.8.3 Activity, consider having students take turns asking clarifying questions and agreeing or disagreeing with their partner or group to encourage discourse.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Throughout this lesson, students are expected to translate fluidly between different representations of proportional relationships in context. Encourage students to continue making connections between each representation.

DIFFERENTIATE INSTRUCTION

This lesson encourages differentiation through a variety of means. Consider using data gathered in previous lessons during this unit to group students purposefully. It may be beneficial to provide additional support to students who are still struggling with unit rates and proportional relationships, so group these students together to complete the activities as a small group with support from the teacher, where appropriate. Alternatively, consider grouping students heterogeneously and positioning the teacher to support with specific stations based on how the flow of each rotation is set up in the classroom.

11.8.1 WARM-UP: CRITICAL THINKING – SHOWER OR BATH?

DIFFERENTIATE INSTRUCTION

Consider breaking the class into six small groups using your formative assessments from previous lessons. Then students can rotate through one set of three stations as directed.



11.8.1 Warm-Up: Critical Thinking – Shower or Bath?

Some people say that taking a bath uses more water than a shower. Others disagree.

1. What information would you collect in order to answer the question of which uses more water?

Answers will vary.

Some information I would collect in order to answer this question is the total number of gallons of water used for a bath versus a shower. Another piece of information I would collect is how long the person was showering for. I would also want to know how many gallons of water the tub holds.

2. What are some proportional relationships that may be involved in helping to answer this question?

Answers will vary.

Gallons used per minute

Gallons per bath

Gallons per shower



11.8.2 Activity: Station 1 – Gym Membership Plans

In January, Gigi signed up for a membership at Total Fitness. The plan she chose costs \$65 in start-up fees and then \$20 per month starting in February. Julien also signed up at Total Fitness in January. His plan costs \$32 per month starting in February, and his start-up fees and January costs were waived.

- Complete the tables to represent the total costs of their gym memberships for both Gigi and Julien for the first 8 months of their memberships.

Gigi's Membership Costs	
Months Since January	Total Cost
0	65
1	85
2	105
3	125
4	145
5	165
6	185
7	205

Julien's Membership Costs	
Months Since January	Total Cost
0	0
1	32
2	64
3	96
4	128
5	160
6	192
7	224

- Determine whether either or both gym memberships are a proportional relationship. Then, complete the statements.

Gigi's membership costs ☐ represent ☒ do not represent a

proportional relationship because Answers will vary. the ratio of months since January to the total cost is not equivalent for all months.

Julien's membership costs ☒ represent ☐ do not represent a

proportional relationship because Answers will vary. the ratio of months since January to the total cost is equivalent for all months.

- For the relationship(s) that is proportional, write an equation to represent the total cost, c , in dollars, for m months of the membership.
 $c = 32m$
- Discuss with your partner which membership seems to be a better deal. Then, summarize your discussion.
Answers will vary.
I think it depends how long they plan on having the membership. If they keep the membership for more than 5 months past January, then Gigi's deal is better because she will pay less per month so it will eventually cost less overall.

11.8.2 ACTIVITY: STATION 1 – GYM MEMBERSHIP PLANS

TEACHER MOVES

Ensure students understand that zero months since January represents the startup fees. This will be important in being able to determine which is the proportional relationship and total costs to each gym member.

TEACHER MOVES

Specifically have students make connections to the table of values in question 1. If students have not made the connection that the length of the membership determines which deal is better, consider prompting them to add more values to each table to highlight this idea.

11.8.3 ACTIVITY: STATION 2 –WHO EARNS MORE MONEY?

DIFFERENTIATE INSTRUCTION

Consider prompting students to create a table of values for Kahlil's wages to allow students to compare the values visually, if needed.

Students could also add a row below the given table to calculate the money Kahlil earns for the same number of hours worked.



11.8.3 Activity: Station 2 – Who Earns More Money?

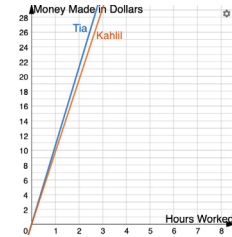
Tia works at the Boys and Girls Club after-school program at Lockhart Middle School in Orlando, FL. The table shows how much money she earned each day she worked last week.

Time Worked (hours)	1.75	2.5	4.25
Money Earned (\$)	18.55	26.50	45.05

Kahlil has a job bagging groceries at Publix where he earns \$9.75 per hour.

- Who would make more money if they both work 10 hours? Explain or show work.

Tia would make more money because she is making $\frac{45.05}{4.25}$ or \$10.60 per hour while Kahlil is only making \$9.75 per hour.



- On the coordinate grid shown, draw a graph that represents y , the amount of money, in dollars, Tia makes for working x hours. Label both axes appropriately.
- On the same grid, draw a graph that represents y , the amount of money, in dollars, Kahlil makes for working x hours.

TEACHER MOVES

Students may struggle to determine which student has the greatest hourly rate of pay. Use scaffolded questions to guide student thinking toward making the connection between the graphs and rates of pay: What does the steepness of the graph say about that person's rate of pay? What is the constant of proportionality of each graph? How is the constant of proportionality related to the rate of pay? What is the value of y on each graph when $x=1$?

- Label each graph with the name of the person whose earnings it represents.
- Explain how the graph shows which person has the greatest hourly rate of pay.
Tia's line is steeper and can be seen above Kahlil's line which helps us see that she has the greatest hourly rate of pay. The value of y when $x = 1$ is also greater on Tia's graph than Kahlil's.
- For each person, determine the constant of proportionality and write an equation to represent the relationship between their total earnings, y , in dollars, and the hours worked, x .

Tia's Earnings (\$)	Constant of Proportionality	10.60
	Equation	$y = 10.60x$

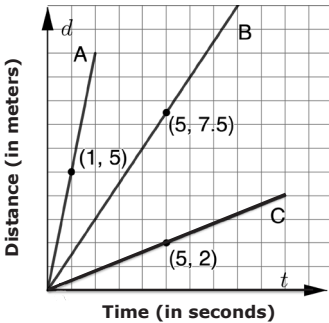
Kahlil's Earnings (\$)	Constant of Proportionality	9.75
	Equation	$y = 9.75x$



11.8.4 Activity: Station 3 – Robot Races

The students in the STEM Club at Umatilla Middle School built solar-powered robots which they raced in the gym at their STEM Showcase.

The graphs shown represent the distance d , in meters, that each of the three robots A, B, and C traveled after t seconds.



1. Determine whether each statement in the table is true or false.

Statement	True or False?
Robot B traveled in a different direction than the other robots.	☐ true ☒ false
Robot B traveled 5 meters in 7.5 seconds.	☐ true ☒ false
Robot B traveled 7.5 meters in 5 seconds.	☐ true ☒ false
Robot B traveled $\frac{2}{3}$ meters per second.	☐ true ☒ false
Robot B traveled $\frac{3}{2}$ meters per second.	☐ true ☒ false

11.8.4 ACTIVITY: STATION 3 – ROBOT RACES

TEACHER MOVES

Position students to reason about each statement before selecting true or false. Encourage mathematical discourse as students use the graphs to determine whether each statement is true or not. Students should not need to complete any computations. Rather, they should use the graphs to reason about each statement.

11.8.4 ACTIVITY: STATION 3 – ROBOT RACES (CONTINUED)

QUESTIONING

To expand students' thinking, consider having students find:

- The distance traveled in meters per minute.
- The distance each robot could travel in one day.

2. Determine the constant of proportionality, $\frac{d}{t}$, for each robot.

Robot	Constant of Proportionality
A	5
B	$\frac{3}{2}$
C	$\frac{2}{5}$

3. Explain what the constant of proportionality determined for each robot represents in this context.

Robot A travels 5 meters per second.

Robot B travels 1.5 meters per second.

Robot C travels $\frac{2}{5}$ of a meter per second.

4. Write an equation to represent the relationship between distance, d , and time, t , for each robot.

Robot	Equation
A	$d = 5t$
B	$d = \frac{3}{2}t$
C	$d = \frac{2}{5}t$

5. Assume the robots traveled at the same rate for 10 seconds. How far would each robot have traveled in that time?

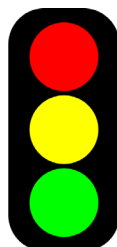
Robot	Distance Traveled in 10 seconds (meters)
A	50 meters
B	15 meters
C	4 meters

**11.8.5 Wrap-Up: Reflection**

So far in this unit, you have learned multiple concepts about proportional relationships including those listed below.

- Determining if two quantities have a proportional relationship
- Determining the constant of proportionality of a proportional relationship given a table, graph, or written description
- Graphing proportional relationships from a table, equation, or written description
- Translating between different representations of proportional relationships

1. Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.



I still feel stuck about how to...

Answers will vary.

I'm starting to get the hang of...

Answers will vary.

I feel confident and ready to go with...

Answers will vary.

11.8.5 WRAP-UP: REFLECTION**DIFFERENTIATE INSTRUCTION**

Consider using students' responses to the 11.8.5 Wrap-Up to form groups for Lesson 9.